

VANIA VELDA

AI Engineer, Speech and Language

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SUMMARY

AI engineer working on speech, audio and machine translation, with 3 years as an AI engineer and 2 in research. As the only AI engineer at a Seoul startup, I build production systems on open-source and open-weight models. My work is used by 500K+ app users, a Korean news broadcaster, a university and a major mobile carrier.

EXPERIENCE

AI Engineer, STRA, Seoul

Aug 2023 – Present

Only AI engineer at the company; I own every model from data to production.

- Built the AI behind KpopPro's lyrics sync and AI covers. The app reached 500K+ downloads and was later acquired.
- Extended lyrics alignment from English to Korean by building a Korean G2P pipeline and fine-tuning a multi-task CTC model: 94.9% accuracy (pCO at 500 ms) in Korean vs 95.2% in English.
- Built Korean-to-English translation for a Korean news broadcaster's live subtitles, raising BLEU on a news test set from 16.5 to 30.8.
- Subtitled and dubbed 7 courses for a Korean university (100+ hours) into English, in each lecturer's cloned voice.
- Built a dubbing platform supporting 20+ languages, with automatic voice matching and quality checks.
- Built AI soccer subtitles for a sports media client: about 500 videos (400+ hours), with professional editors keeping over 90% of the AI translation.
- Built source separation that raised speech recognition accuracy on noisy audio from 91% (raw) to 96%, helping win a contract with a major Korean mobile carrier.
- On a medical interpreting test set, glossary-constrained training raised medical term accuracy from 43.9% to 80.3%.

Graduate Researcher, Vision Language Intelligence Lab, Sejong University

Sep 2021 – Aug 2023

- Video captioning with Transformers and CLIP: first-author journal paper and a Korean patent application. TA for machine learning and AI courses.

Software Developer, freelance and contract roles, Indonesia

2019 – 2021

- Built LMS integrations, a COVID-19 tracing website, and Android and web apps.

PUBLICATIONS

- STRA-MT: Deterministic Terminology Control for Document-Level Machine Translation. WMT 2026 (accepted). First author.
- What Transfers and What Counts: Terminology Inflection in Multilingual NMT. NLPPIR 2026 (accepted). First author.
- Improving Distinctiveness in Video Captioning with Text-Video Similarity. Image and Vision Computing, 2023. First author.
- Action Knowledge for Video Captioning with Graph Neural Networks. JKUSU-CIS, 2023. Co-author.

SKILLS

Speech and audio: ASR, TTS, voice cloning, voice conversion, source separation, forced alignment

Language and LLMs: machine translation, LLM post-editing, terminology control, RAG

Tools: Python, PyTorch, Hugging Face, FastAPI, Docker, Redis, OpenAI API, Gemini API, ElevenLabs API

Languages: English (fluent), Indonesian (native), Chinese (intermediate)

EDUCATION

M.S., Intelligent Mechatronics Engineering, Sejong University, Seoul

2021 – 2023

B.S., Informatics (Computer Science), Institut Teknologi Bandung, Indonesia

2017 – 2021

Awarded the JDF Scholarship for academic excellence (up to 3 recipients per year).